



TF-A: Trusted Firmware



Concept of FIP

- ▶ FIP = *Firmware Image Package*
- ▶ Concept specific to TF-A
- ▶ *packaging format used by TF-A to package firmware images in a single binary*
- ▶ Typically used to bundle the BL33, i.e. the U-Boot bootloader that will be loaded by TF-A.
- ▶ https://trustedfirmware-a.readthedocs.io/en/latest/getting_started/tools-build.html
- ▶ https://wiki.st.com/stm32mpu/wiki/How_to_configure_TF-A_FIP



Configuring TF-A

- ▶ TF-A does not use *Kconfig* for configuration
- ▶ All the configuration is based on variables passed on the `make` command line
- ▶ Most variables are documented at: https://trustedfirmware-a.readthedocs.io/en/latest/getting_started/build-options.html



Configure TF-A: important variables

- ▶ `CROSS_COMPILE`, cross-compiler prefix
- ▶ `ARCH`, CPU architecture: `aarch32` or `aarch64`
- ▶ `ARM_ARCH_MAJOR`, `7` for ARMv7, `8` for ARMv8
- ▶ `PLAT`, SoC family, any directory name in `plat` that contains `platform.mk`
- ▶ `AARCH32_SP`, the Secure Payload, specific to ARMv7. Either OP-TEE or the built-in *SP-MIN* provided by TF-A
- ▶ `DTB_FILE_NAME`, path to the Device Tree describing our board
- ▶ `BL33`, path to the second stage bootloader, usually U-Boot, to include in the FIP image
- ▶ Specific to STM32MP1
 - `BL33_CFG`, path to the U-Boot Device Tree
 - `STM32MP_SDMMC=1`, enable support for SD card/eMMC in TF-A



Building TF-A for STM32MP1

```
\$_make CROSS_COMPILE=arm-linux-  
  ARM_ARCH_MAJOR=7  
  ARCH=aarch32  
  PLAT=stm32mp1  
  AARCH32_SP=sp_min  
  DTB_FILE_NAME=stm32mp157a-dk1.dtb  
  BL33=/path/to/u-boot/u-boot-nodtb.bin  
  BL33_CFG=/path/to/u-boot/u-boot.dtb  
  STM32MP_SDMMC=1  
  fip all
```

Build results in `build/stm32mp1/release`. Important files:

- ▶ `tf-a-stm32mp157a-dk1.stm32`, TF-A itself
- ▶ `fip.bin`, the FIP image, containing U-Boot and other elements



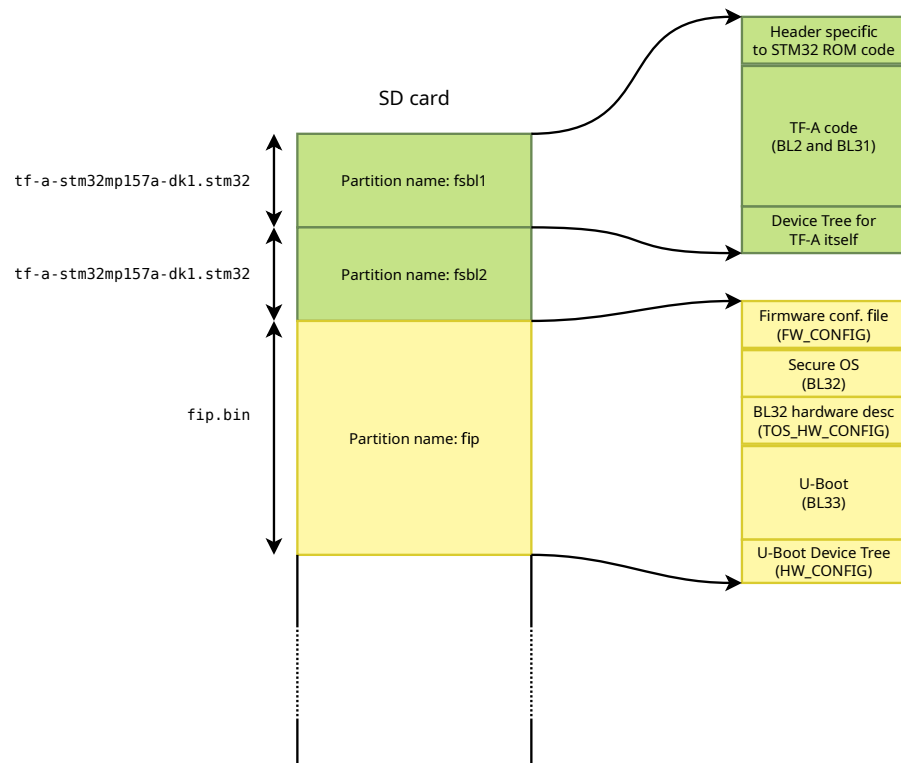
FIP image contents

fiptool info

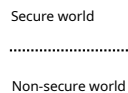
```
\$ \_. /tools/fiptool/fiptool info build/stm32mp1/release/fip.bin Secure Payload BL32 (Trusted OS): offset=0x100,
size=0x8AEC, cmdline="--tos-fw"
Non-Trusted Firmware BL33: offset=0x8BEC, size=0xECE6C, cmdline="--nt-fw"
FW_CONFIG: offset=0xF5A58, size=0x226, cmdline="--fw-config"
HW_CONFIG: offset=0xF5C7E, size=0x16A98, cmdline="--hw-config"
TOS_FW_CONFIG: offset=0x10C716, size=0x3CF6, cmdline="--tos-fw-config"
```



STM32MP1 partition layout

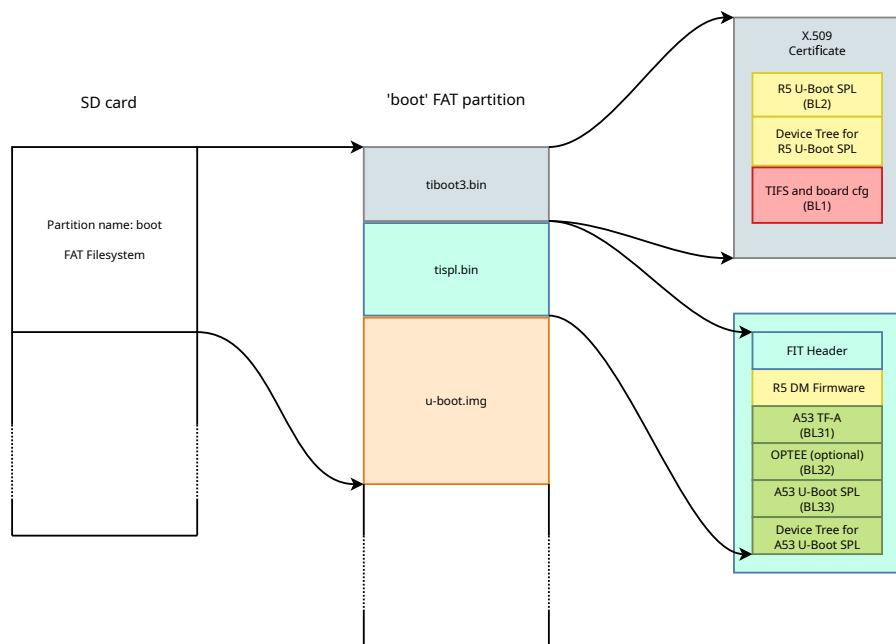


Reminder: boot sequence with TF-A on STM32MP1





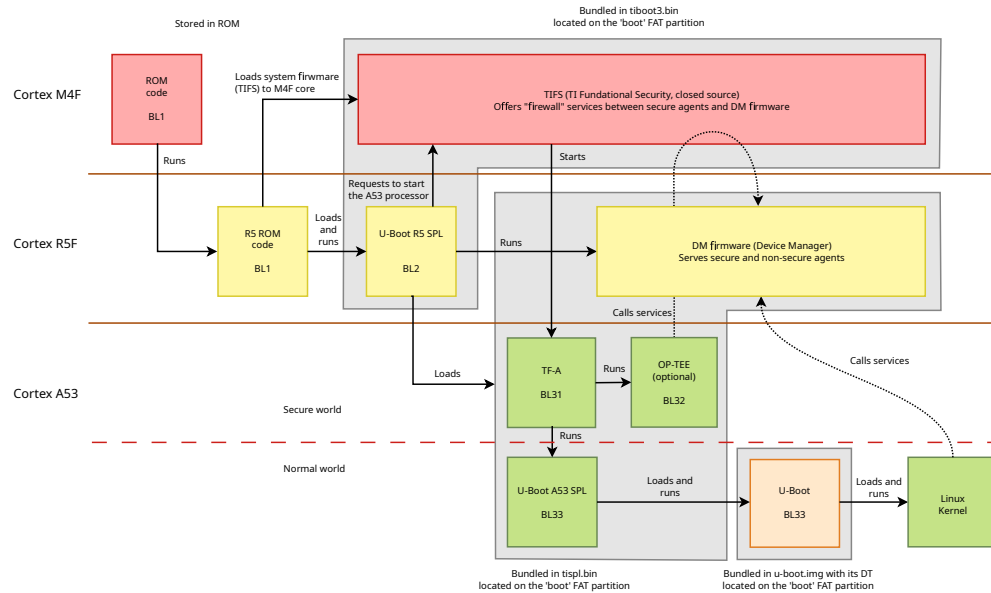
AM62x (BeaglePlay) partition layout



Reminder: boot sequence with TF-A on
AM62x



AM62x (BeaglePlay) partition layout



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